ELECTRICAL MACHINES · PAPER

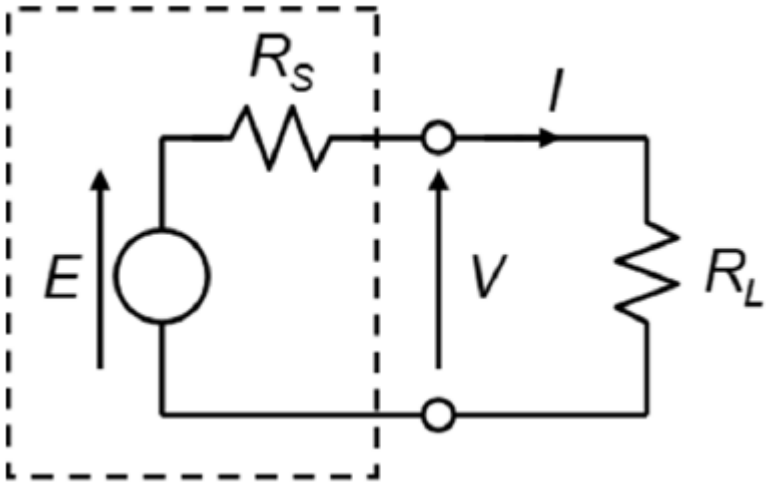
Jun 2023 exam: full worked solutions

Practice questions · Day 18 · Tue 10 Nov · pack MX1

Question 1. AC and DC Power Circuit Analysis [20 marks]

1a) Battery Modelling [5 marks]

The equivalent circuit on the right is used to model a car battery.

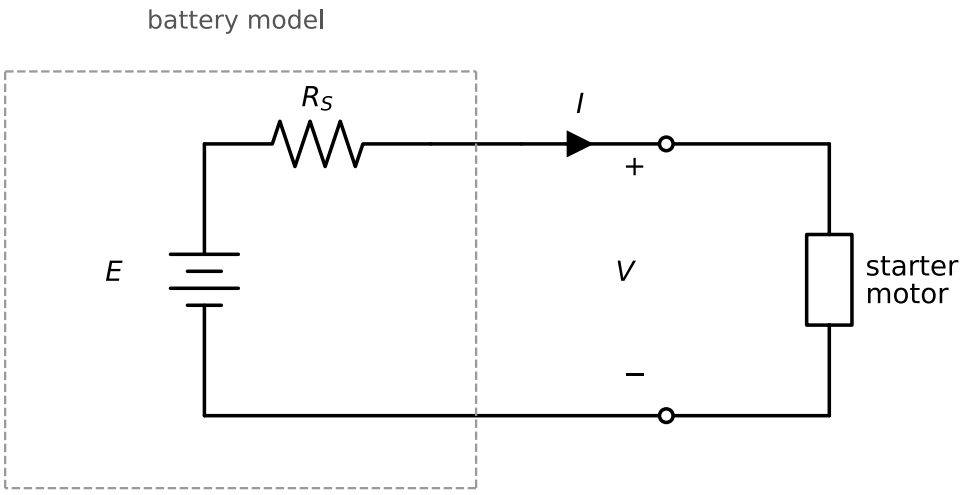


Jun23 Q1a figure: battery model, E in series with R_S, terminal voltage V, load R_L (from the exam paper)

i) The battery's open-circuit voltage is measured at 13.8 V. Find the value of the internal voltage source E. [1 mark]

ii) A DC starter motor is connected to the battery. It draws a current of 300 A and the car battery voltage falls to 9.5V. Find the value of the internal resistance of the battery, R_S, the power dissipated in this resistance, and the power delivered to the motor. [3 marks]

iii) Find the battery output voltage when a load drawing a current of 20A is connected to it. [1 mark]



open circuit: V = 13.8 V; at I = 300 A: V = 9.5 V

Q1a: battery modelled as EMF E in series with RS, feeding the starter motor; open circuit 13.8 V, 9.5 V at 300 A

MARKS: (i) 1. (ii) 1 for R_S, 1 for the loss, 1 for the motor power. (iii) 1.

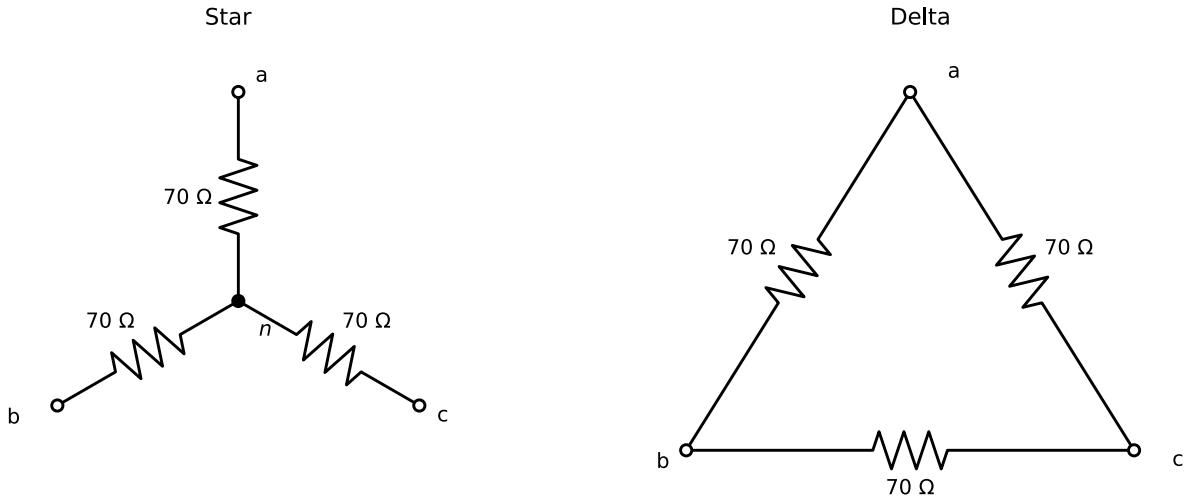
> **COMMON MISTAKE:** giving $EI = 4140$ W as the motor power. The power delivered to the load uses the **terminal** voltage, VI.

1b) Star-Delta Equivalent Circuits [6 marks]

Consider a three-phase load consisting of three 70 Ω resistors connected to a 460V, 3ph supply.

i) If the load is star-connected, calculate the magnitudes of the phase current, line current and total power. [3 marks]

ii) If the load is delta-connected, calculate the magnitudes of the phase current, line current and total power. [3 marks]



line voltage 460 V between any two of a, b, c

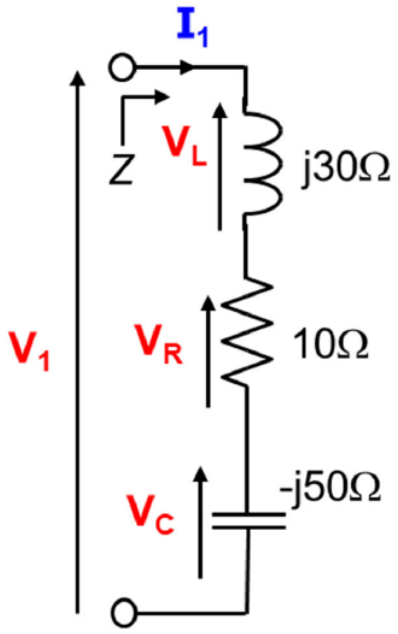
Q1b: three 70 ohm resistors in star and in delta on a 460 V line supply

MARKS: 1 each for phase current, line current and power, in each case (3 + 3).

> **COMMON MISTAKE:** using 460 V across each resistor in star (gives 3 times the true power), or forgetting $\sqrt{3}$ between phase and line current in delta.

1c) Phasor Calculations [9 marks]

Consider the single-phase series RLC load as shown in the figure on the right. The current $\mathbf{I}_1$ is known to be $13\angle+20^\circ$ A.



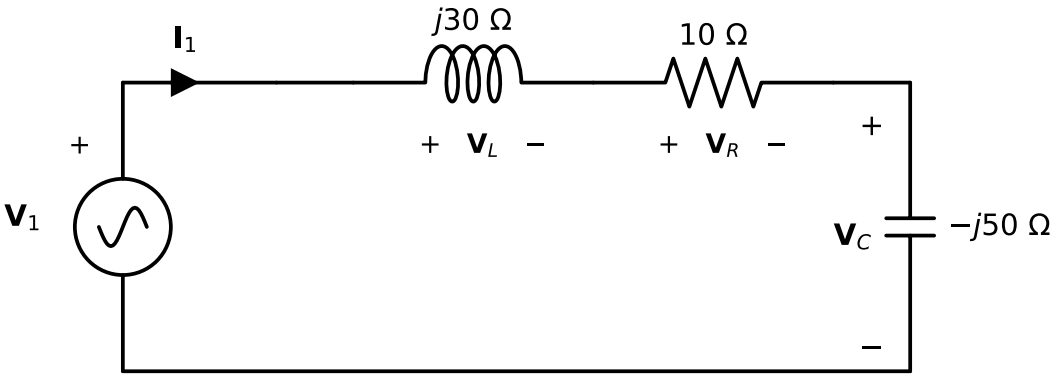
Jun23 Q1c figure: series j30 Ω, 10 Ω and –j50 Ω carrying I1, with V_L, V_R, V_C and V1 marked (from the exam paper)

i) Using the known current $\mathbf{I}_1$, find the voltage phasors across the resistance $\mathbf{V}_R$, inductance $\mathbf{V}_L$ and capacitance $\mathbf{V}_C$ in polar form using Ohm's law. [3 marks]

ii) Find the phase impedance Z of the load in polar form and hence the voltage phasor $\mathbf{V}_1$ in polar form using Ohm's law. [3 marks]

iii) The complex power S is given by $S = \mathbf{V}_1\mathbf{I}_1^*$, where * refers to the complex conjugate. Find the complex conjugate of the current $\mathbf{I}_1^*$, thus the complex power S and hence the total real power P and reactive power Q supplied from the power source. [3 marks]

$\mathbf{I}_1 = 13\angle 20^\circ$ A (element values from the paper's figure, as in pack M1)



Q1c: series j30 ohm, 10 ohm and -j50 ohm carrying I1 = 13 at 20 degrees A (values from the paper's figure, as in pack M1)

MARKS: (i) 1 per voltage. (ii) 1.5 for Z, 1.5 for $\mathbf{V}_1$. (iii) 1 for $\mathbf{I}_1^*$, 1 for S, 1 for P and Q with signs.

Discrepancies with the official sheet:

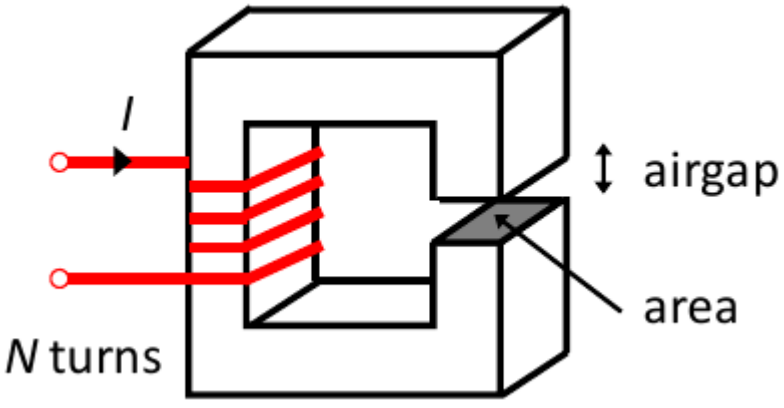
- The official sheet gives $\mathbf{V}_R = 130\angle 120^\circ$ V. This is a **typo**: a resistor's voltage is in phase with its current, so it is $130\angle 20^\circ$ V. The official $\mathbf{V}_1$ (which needs $\mathbf{V}_R$ at 20°) confirms this.
- The official Q is -3379 VAr. The exact value is -3380 VAr ($13^2 \times 20$). The official number comes from multiplying the rounded 290.7 V. Either earns the mark.

> **COMMON MISTAKE:** writing $\mathbf{V}_C$ at $+110^\circ$ or adding the reactances as $30 + 50$. A capacitor is $-jX_C$, so its voltage lags the current by 90°.

Question 2. Magnetic Circuits and Transformers [20 marks]

2a) Electromagnetics [9 marks]

Consider the iron c-core shown, which has a magnetic cross-sectional area of 20cm² and a mean magnetic path length of 50cm. The core has a winding with 32 turns and has a relative permeability of 3,000. The airgap length is 1.5mm.

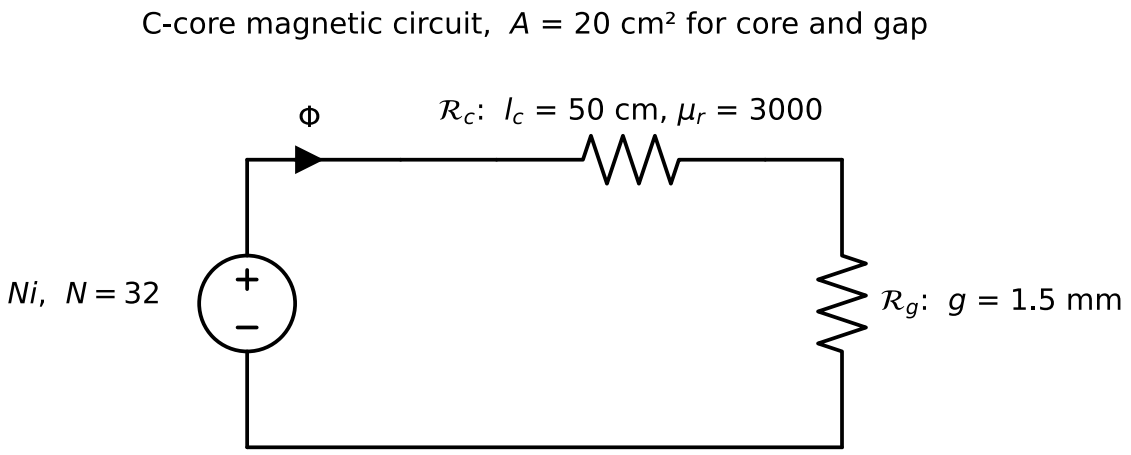


Jun23 Q2a figure: iron c-core with an N-turn winding carrying I, and an airgap (from the exam paper)

i) Calculate the reluctances of the airgap $\mathcal{R}_G$ and the core $\mathcal{R}_C$ and hence find the inductance of the coil. [4 marks]

ii) Calculate the maximum energy stored in the airgap only, if the maximum allowable peak core flux density is 1.8 T. [2 marks]

iii) The magnetic core is operating with a peak flux density of 1.5T at 60 Hz. It has an iron loss of 12W, of which 30% is hysteresis losses and 70% is eddy-current losses. Calculate the new hysteresis, eddy-current and total iron loss if the peak flux density is reduced to 1.2T, whilst the frequency is increased to 100Hz. [3 marks]



Q2a: magnetic circuit, MMF Ni with N = 32 driving flux through the core and gap reluctances in series

MARKS: (i) 1 each for $\mathcal{R}_G$ and $\mathcal{R}_C$, 2 for L (formula and value). (ii) 1 formula, 1 value. (iii) 1 for the split, 1 for each scaled loss (and the total).

COMMON MISTAKE: in (iii), scaling the whole 12 W by one law. In (ii), using $\frac{1}{2}LI^2$ (total energy) when the question asks for the gap only.

2b) Transformer Parameters [5 marks]

The following tests are performed on a single-phase 240V:32V, 2400VA transformer.

- Short-circuit test. Primary measurements are 27 V, 9.7 A, and 55 W.
- Open-circuit test. Primary measurements are 240 V, 1.3 A, and 45 W.

Find all the simplified equivalent circuit parameters.

Q2b test circuits: OC 240 V, 1.3 A, 45 W sees only the shunt branch; SC 27 V, 9.7 A, 55 W sees only the series branch

MARKS: 1 for each of R, X, R_C, X_M, plus 1 for stating they are referred to the primary.

COMMON MISTAKE: $X_M = V/I_{OC} = 184.6\ \Omega$. That is $|Z_M|$ of the parallel pair, not X_M. Separate P and Q first, or find the magnetising current $I_M = I_{OC}\sin\phi$.

2c) Transformer Analysis [6 marks]

Consider the transformer simplified equivalent circuit shown on the right. The values of the primary voltage $\mathbf{V}_1$ and current $\mathbf{I}_1$ phasors are known and it is desired to find the value of the secondary voltage $\mathbf{V}_2'$ phasor.

Jun23 Q2c figure: simplified transformer equivalent circuit, R_C parallel jX_M across V1, then jX1+jX2′ and R1+R2′ to V2′ (from the exam paper)

i) Using Kirchhoff's voltage law (KVL), write an expression for $\mathbf{V}_2'$ in terms of $\mathbf{V}_1$, $\mathbf{I}_2'$ and the relevant equivalent circuit parameters shown. [2 marks]

ii) Assuming the load is purely resistive with a value of R_L', use the voltage divider rule to write an expression for $\mathbf{V}_2'$ in terms of $\mathbf{V}_1$ and the relevant equivalent circuit parameters shown. [2 marks]

iii) For a purely resistive load, would you expect that $\mathbf{V}_2'$ will be leading, in phase or lagging with respect to $\mathbf{V}_1$? Carefully explain why. [2 marks]

Q2c: simplified transformer circuit, shunt RC parallel jXM across V1, series R1 + R2′ and j(X1 + X2′) to the load RL′

MARKS: (i) 2 (correct sign and both series terms). (ii) 2. (iii) 1 for "lagging", 1 for the reason chain (current lags because of X; load voltage in phase with current).

COMMON MISTAKE: in (iii), saying "lagging because it is a transformer" with no reason, or saying "in phase because the load is resistive". The load voltage is in phase with $\mathbf{I}_2'$, not with $\mathbf{V}_1$.

Question 3. AC and Induction Machines [20 marks]

3a) AC Machines [6 marks]

Consider a 50 Hz AC machine which has a rated speed of 560 rpm.

i) How many poles does it have? [1 mark]

ii) Is the AC machine a synchronous machine or an induction machine? Briefly explain why. [1 mark]

iii) At rated speed, what is the ideal (that is, maximum possible) efficiency of the machine? [2 marks]

iv) The machine is operating at one-quarter of rated torque. What is the operating speed in rpm? You can assume a linear torque versus speed curve. Find also the output power expressed as a percentage of rated output power. [2 marks]

MARKS: (i) 1. (ii) 1 (answer with reason). (iii) 1 for $\eta = 1 - s$, 1 for the value. (iv) 1 for 590 rpm, 1 for 26.34%.

> **COMMON MISTAKE:** in (iv), saying the power is 25%. The torque is a quarter, but the speed went up slightly, so the power is a little more than a quarter.

3b) Induction Machine Torque vs Speed [5 marks]

The equivalent-circuit parameters of a wound-rotor induction machine are listed below.

stator resistance R_1 = 0.4 Ω, rotor resistance R_2' = 0.5 Ω
stator leakage reactance X_1 = 1.3 Ω, rotor leakage reactance X_2' = 1.5 Ω
magnetising reactance X_M = 95 Ω, core loss resistance R_C = 270 Ω

i) Calculate the external resistance, R_{EXT}, required such that maximum torque occurs at standstill. [2 marks]

ii) On the axes below, draw a typical torque vs. speed characteristic curves for an induction machine – assume that this curve represents the above induction machine with zero external

resistance. In addition, draw the torque curves for external resistance equal to 0.5 and 1pu, where 1pu represents the R_{EXT} calculated in part i) above. [3 marks]

(The paper gives blank torque–speed axes, torque up and speed n across, starting at 0.)

Q3b: simplified circuit with 270 ohm parallel j95 ohm, 0.4 ohm, j1.3 ohm, j1.5 ohm and (R2′ + REXT′)/s

MARKS: (i) 1 for the s_{max} condition, 1 for the value. (ii) 1 for the correct $R_{EXT} = 0$ shape (zero at n_S, peak near n_S, lower starting torque), 1 for equal peaks, 1 for the 1 pu curve peaking at $n = 0$.

> **COMMON MISTAKE:** drawing the curves with different peak heights, or making them cross zero at different speeds. Adding rotor resistance only stretches the curve along the slip axis.

3c) Induction Machine Torque Calculation [9 marks]

A 415 V, 50 Hz, 1450 rpm, three-phase, star-connected induction machine has the following equivalent circuit parameters (referred to the stator winding where appropriate):

stator resistance R_1 = 0.3 ohms, rotor resistance R_2' = 0.2 ohms
stator leakage reactance X_1 = 2.5 ohms, rotor leakage reactance X_2' = 2.5 ohms
magnetising reactance X_M = 50 ohms, core loss resistance R_C = 200 ohms

In your calculations you should use the simplified equivalent circuit.

i) Find the number of poles and hence the synchronous speed in rpm. [2 marks]

ii) Consider the machine operating at 1520 rpm. Is it operating as a motor or generator? Briefly indicate why? [1 mark]

iii) Consider the machine operating at rated voltage at its rated speed of 1450 rpm. What is the output torque? Hint: use the torque formula. [3 marks]

iv) Consider the machine operating at rated voltage at its rated speed of 1450 rpm. Ignoring the magnetising reactance and core loss resistance, find the input current magnitude and phase. [3 marks]

Q3c: simplified circuit with 200 ohm parallel j50 ohm, 0.3 ohm, j2.5 ohm, j2.5 ohm and R2′/s

MARKS: (i) 1 poles, 1 n_S. (ii) 1 with reason. (iii) 1 for s and ω_S, 1 for the formula with phase voltage, 1 for the value. (iv) 1 for the impedance, 1 for the magnitude, 1 for the angle.

COMMON MISTAKE: using the line voltage 415 V in the torque formula (gives 3 times the torque, 305 Nm), or dividing by the rotor speed ω instead of ω_S. $\tau = P_{AG}/\omega_S$ always.

Question 4. Synchronous Machines [20 marks]

4a) Synchronous Machines [5 marks]

A 150 MVA, 50Hz, 3,000rpm, 14 kV power station generator gives the following test results:

Excitation (amps)	Open-circuit line voltage (kV)	Short-circuit current (A)
60	4.6	720

The stator resistance is measured at 0.3 ohm.

i) Find the value of synchronous reactance in ohms (not per-unit). [2 marks]

ii) The generator is delivering a real power of 100MW and reactive power of 50 MVAr at rated voltage. Calculate the magnitude and phase of the stator voltage and current in per-unit. [3 marks]

Q4a test circuits: OC 4.6 kV line and SC 720 A, both at a field current of 60 A, R = 0.3 ohm

MARKS: (i) 1 for $|Z_S|$ with phase voltage, 1 for X_S. (ii) 1 for $\mathbf{V}$, 1 for the magnitude of $\mathbf{I}$, 1 for the (negative) angle.

Discrepancy: the official sheet gives 0.7453 pu. The exact value is $\sqrt{0.6667^2 + 0.3333^2} = 0.74536$, which rounds to 0.7454. This is just truncation in the

official sheet.

> **COMMON MISTAKE:** a positive current angle (the OCR'd text version of the answer sheet drops the minus sign, but the PDF shows $-26.57°$), or using the line OC voltage directly ($4600/720 = 6.39\ \Omega$, a factor of $\sqrt{3}$ too big).

4b) Power Factor Correction [7 marks]

An industrial plant uses two identical induction motors, each driving a mechanical load of 200kW. These motors operate with efficiencies of 85% and a power factor of 0.83 lag. One of the motors develops a fault and is replaced with a synchronous motor rated at 250 kVA which has an efficiency of 90%.

i) Calculate the input real and reactive power drawn by the remaining induction motor. [2 marks]

ii) Calculate the input real power and the maximum reactive power capability of the synchronous motor when driving the 200 kW load. [2 marks]

iii) Use the above results to determine the maximum power factor for the plant. [3 marks]

Q4b plant: the remaining induction motor and the 250 kVA synchronous motor in parallel on the supply

MARKS: (i) 1 each. (ii) 1 each. (iii) 1 for P total, 1 for net Q with the correct sign, 1 for pf with "lag".

Discrepancy: the official sheet gives 114.6 kVAr. The exact value is 114.53 kVAr. The official number comes from using the rounded 222.2 kW ($\sqrt{250^2 - 222.2^2} = 114.58$). The final pf is 0.9955 either way.

> **COMMON MISTAKE:** using the 200 kW output instead of the input power, adding the two reactive powers instead of subtracting, or treating 250 kVA as real power.

4c) Short-Answer Questions [8 marks]

Answer the following questions using words only (that is, without using any sketches, drawings or equations).

i) Explain the concept of power-factor correction. Why is power-factor correction useful in power systems? [3 marks]

ii) Carefully explain the differences and similarities between transformers and induction machines. [3 marks]

iii) Consider a squirrel-cage induction machine. If the rotor cage was made from a perfect conductor (with infinite conductivity) how would this affect the properties of the machine? [2 marks]

MARKS: (i) 1 what reactive power is and why it is a problem, 1 capacitor in parallel generates reactive power, 1 the benefit (lower current, losses, better use of rating). (ii) about 1.5 for similarities, 1.5 for differences (rotating short-circuited secondary, different rotor frequency, R_2'/s). (iii) 1 for "zero R_2', no rotor loss, higher efficiency", 1 for "vertical τ–n curve at n_S and zero starting torque".

> **COMMON MISTAKE:** in (iii), saying the machine would be "more efficient" and stopping there. The second mark needs the zero starting torque. In all of 4c, do not draw a phasor diagram or write a formula: the question forbids it.